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Can a Neural Network Measure a Critical Temperature?

Finite-size scaling of classifier outputs for the 3D Ising model and the SK spin glass

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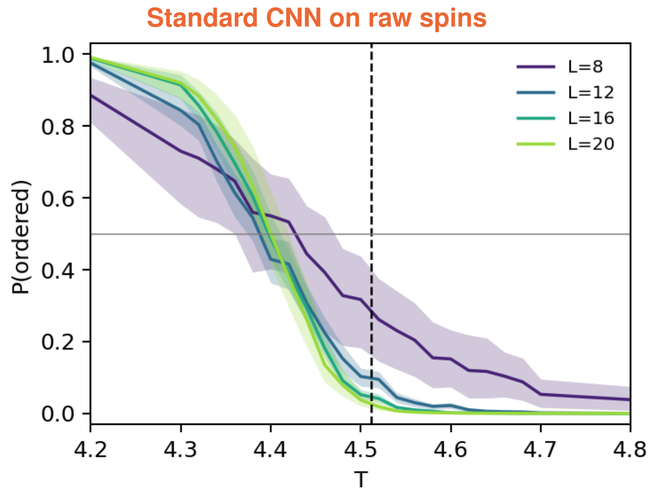
The question

Neural networks are often used to locate phase transitions. They are trained on clearly ordered and clearly disordered samples, and the temperature where the output is $\frac{1}{2}$ is taken as the critical temperature T_c. But $\frac{1}{2}$ only marks where the network is undecided. **Is that a measurement of T_c, or an artefact of the training?**

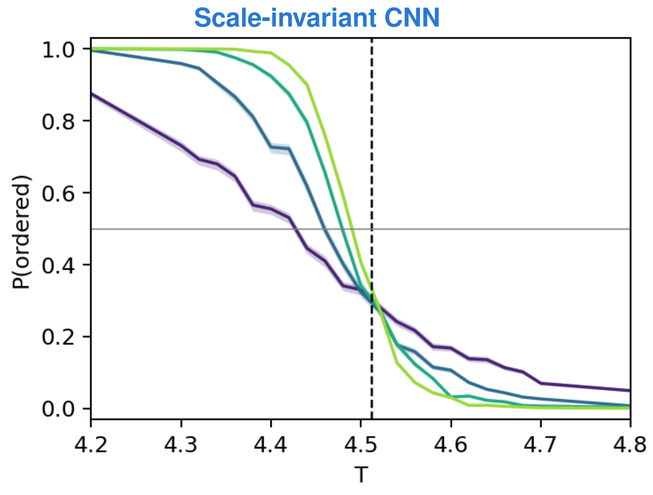
The answer

It is a measurement only if the network sees a quantity that *looks the same at T_c for every system size*. Then T_c is where the output curves for different sizes **cross**, whatever the network learned. Tested on the 3D Ising model (**within 0.01** of T_c) and on a spin glass (**within 0.05**).

01 Same data, same training — only the input differs



Network sees all the raw spins. The curves for different sizes do not meet at one point. Reading $P = \frac{1}{2}$ gives 4.40, **2.5 % too low**.



Network sees one rescaled number. All sizes cross in one point: $\mathbf{4.521(7)}$, within 0.01 of the true $T_c = 4.5115$.

How to read: 3D Ising model. y-axis = the network's confidence that a sample is ordered (magnetic). One curve per lattice size $L = 8, 12, 16, 20$ (dark = small, light = large); bands = spread over five training runs. Dashed line: true T_c; grey line: $P = \frac{1}{2}$.

02 The idea

At T_c the system has no typical length scale.

Zoomed out, it looks the same at every scale, so some quantities have the same statistics for every size L. Example: at T_c the magnetization shrinks as $|m| \propto L^{-\beta/\nu}$, so $|m|\, L^{\beta/\nu}$ is the same for every L.

If the network only sees such a quantity,

$$P(T, L) = F\big((T - T_c)\, L^{1/\nu}\big)$$

and at $T = T_c$ every size gives the same P:
the curves must cross at T_c.

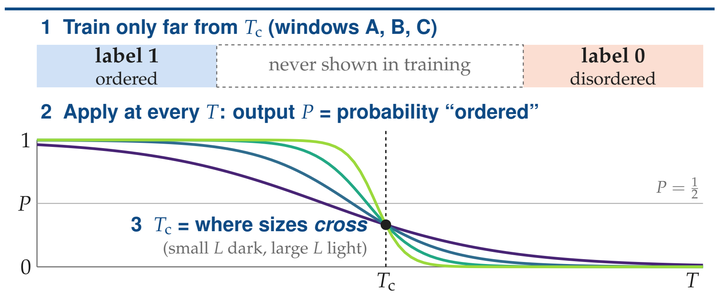
A network on raw spins has no such guarantee: it can use other features, and where it places $P = \frac{1}{2}$ is set by the training data.

Two ways to read T_c from $P(T, L)$

- **Level crossing** (usual): $P = \frac{1}{2}$ at $T^*(L) = T_c(1 + x^* L^{-1/\nu})$; the offset x^* is set by the training, so T_c needs $L \to \infty$.
- **Size crossing** (this work): $F(tL_1^{1/\nu}) = F(tL_2^{1/\nu}) \Rightarrow t = 0$: the curves meet at T_c for *any* weights.

From the RG: $f_s(t, h, L^{-1}) = b^{-d} f_s(b^{1/\nu} t,\, b^{y_h} h,\, bL^{-1})$; with $b = L$, $\langle |m| \rangle = L^{-\beta/\nu} \Phi(tL^{1/\nu})$. 3D Ising: $\nu = 0.630$, $\beta/\nu = 0.518$.

03 How the test works



- **Train far from T_c**: label "ordered" well below, "disordered" well above. The network never sees the region around T_c.
- **Three training windows** (A, B, C) are placed differently on purpose. A physical estimate must *not* depend on the window.
- Every network is trained **five times**; results are compared with the standard non-ML estimator (Binder cumulant $U_4 = 1 - \langle m^4 \rangle / 3\langle m^2 \rangle^2$).

04 The two networks

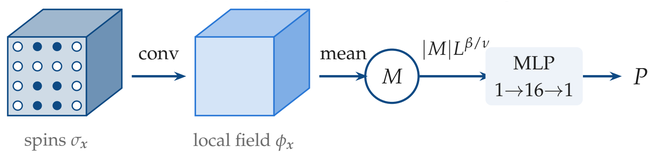
Standard CNN (sees all L^3 spins)

Two 3×3×3 convolutions, 32 channels, periodic padding:

$$h_{c,\mathbf{x}} = \mathrm{ReLU}\Big(\sum_{\mathbf{u}} W_{c,\mathbf{u}}\, \sigma_{\mathbf{x}+\mathbf{u}} + b_c \Big),$$

then a lattice average (works for any L) and an MLP 32→32→1 giving $P = 1/(1 + e^{-z})$.

Scale-invariant CNN (sees one number)

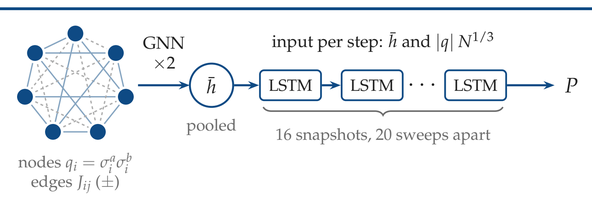


$$\phi_{\boldsymbol{x}} = \sum_c w_c \tanh\Big(\sum_{\boldsymbol{u}} W_{c,\boldsymbol{u}} \sigma_{\boldsymbol{x}+\boldsymbol{u}} \Big),$$

$$M = \frac{1}{N} \sum_{\boldsymbol{x}} \phi_{\boldsymbol{x}}, \qquad M[-\sigma] = -M[\sigma].$$

No biases and an odd tanh make M flip sign with the spins, like the magnetization. Only $|M|L^{\beta/\nu}$ reaches the classifier.

05 Spin-glass network: GNN + LSTM



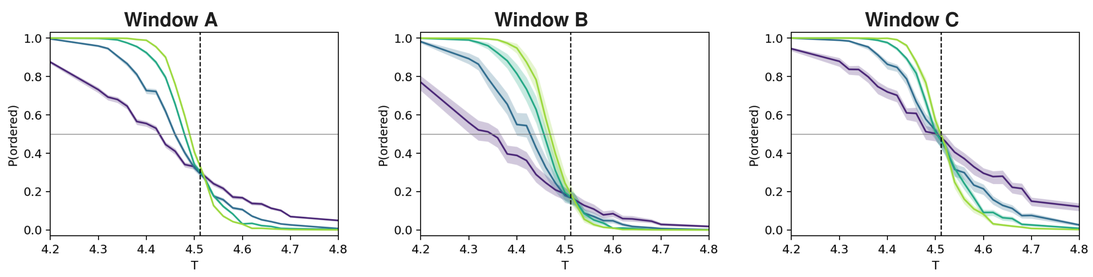
$$h_i^{(k+1)} = h_i^{(k)} + \mathrm{MLP}_k\Big[h_i^{(k)},\ \textstyle\sum_j J_{ij} W_k h_j^{(k)} \Big]$$

Message passing on the complete graph with the couplings J_{ij} as edge weights (2 layers, 32 dim.), averaged to $\bar{h}$; an LSTM reads 16 consecutive snapshots. Compared with *overlap only* (LSTM on $|q|N^{1/3}$) and *single replica* (node feature σ_i^a).

06 Training and error bars

- Label 1 for $T \le T_{\mathrm{low}}$, 0 for $T \ge T_{\mathrm{high}}$ (four temperatures each); inputs multiplied by a random global sign.
- Binary cross-entropy, Adam (lr 10^{-3}, weight decay 10^{-4}), PyTorch; held-out data only.
- Error bars: bootstrap (300 resamples) combined with the spread over five training runs.

07 Result 1, 3D Ising: the *height* moves, not T_c



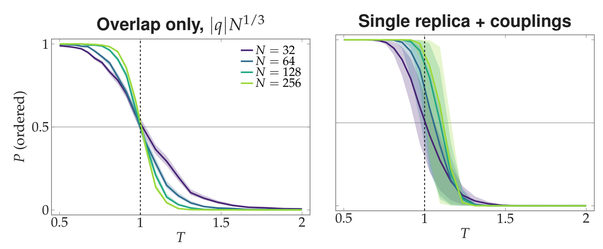
Scale-invariant CNN, training windows A, B, C: the curves meet at heights 0.30, 0.17, 0.48, but always at the dashed line T_c.

Estimate of T_c		true value 4.5115
Standard CNN, $P = \frac{1}{2}$	changes with window	4.40–4.44
Scale-inv. CNN, crossing	same for all windows	**4.518–4.521**
Binder cumulant (no ML)	standard method	4.522

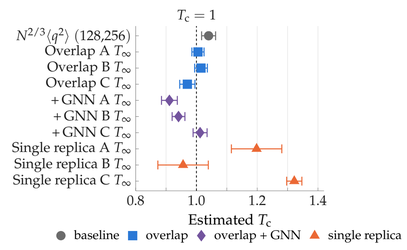
Why only the height moves: training fixes the learned F and hence the height $F(0)$; the temperature $t = 0$ is fixed by the physics. With a wrong $\beta/\nu = 0.1$ the crossing shifts by ≈2 % and the window dependence returns.

08 Result 2: a spin glass (SK model, exact $T_c = 1$)

In a spin glass the couplings are **random**: no magnetization to look at. **Spins alone are useless**: by an exact symmetry, every snapshot is equally likely at every T, and a spins-only network scores 48–51 % (**chance**). **Fix:** run two copies with the same couplings; their **overlap** $q = \frac{1}{N} \sum_i \sigma_i^a \sigma_i^b$ plays the role of m, and the network gets $|q|\, N^{1/3}$.



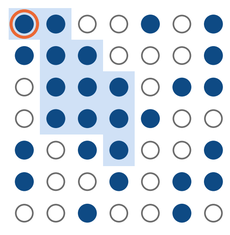
Left: overlap input; $N = 32$–256 cross at $T_c = 1$. **Right:** one copy + couplings: the network measures the temperature (via the energy), not the transition.



Overlap: within ±0.05 of 1 for every window. Single copy: 0.92–1.32. Learned graph features (purple) break the guarantee.

09 Simulation details: Monte Carlo sampling

Wolff clusters (Ising, $L = 8$–20)



Add parallel neighbours with $p_{\mathrm{add}} = 1 - e^{-2J/T}$, flip the cluster. With m (n) boundary bonds not added before (after):

$$\frac{W(\sigma \to \sigma')}{W(\sigma' \to \sigma)} = (1 - p_{\mathrm{add}})^{m-n} = e^{-\Delta E/T}.$$

Parallel tempering (spin glass, $N = 32$–256)

Replicas at $T_1 < \cdots < T_K$ swap configurations to escape the rugged landscape:

$$P_{\mathrm{swap}} = \min\Big(1,\ e^{(\beta_k - \beta_{k+1})(E_k - E_{k+1})}\Big).$$

Ising: one Metropolis sweep + 5 Wolff cluster flips per step (removes critical slowing down), 1000 samples per (L, T), $L = 8$–20. SK: parallel tempering with two replica ladders per disorder sample, $N = 32$–256; equilibration checked with the exact energy–overlap identity $e = -(1 - [\langle q^2 \rangle])/2T$.

Take-home

1. Read T_c where the curves for different system sizes **cross**, not where $P = \frac{1}{2}$.

2. This works only if the network's input **looks the same at T_c for every size**, in a magnet and in a spin glass.

3. Then the network matches, but does not beat, standard finite-size scaling, and the exponent must be supplied.

J. Carrasquilla, R. G. Melko, Nat. Phys. **13**, 431 (2017) · D. Kim, D.-H. Kim, PRE **98**, 022138 (2018) · V. Chertenkov *et al.*, PRE **108**, L032102 (2023) · H. Munoz-Bauza *et al.*, J. Stat. Mech. 073302 (2020) · U. Wolff, PRL **62**, 361 (1989) · K. Hukushima, K. Nemoto, JPSJ **65**, 1604 (1996) · A. M. Ferrenberg *et al.*, PRE **97**, 043301 (2018) · D. Sherrington, S. Kirkpatrick, PRL **35**, 1792 (1975).

Code: github.com/isingchowon/nn-critical-temperature

Code & data